

Neurotech Hub – Practical

Congratulations on making it to the practical phase! This is an open-resource exercise (use forums, AI, etc.) to generally assess your technical capabilities. Please attempt each problem; each should take ~1 hour and you may stop at any point. You will be asked to demonstrate a specific understanding of your work. Questions will not be fielded—interpret and complete the problem as best you can.

Reply by May 1, 2025.

(A) Low Power Datalogging

You are designing a low power data logging circuit. A microcontroller should wakeup every 10 minutes, write 5 rows of data to a uniquely named CSV file on a micro-SD card, then go back to sleep.

1. Design a low power, space constrained sub-circuit that can disconnect power from the SD card during sleep using a single GPIO.
2. Choose an Arduino IDE compatible microcontroller and write the minimum amount of code for this application. Comment as needed.
3. Estimate the current draw during each phase of your program's operation.
4. Add a readme to your project to help someone use your code and indicate what tools you used in your development (IDEs, AI, etc.). Push the entire project to a public GitHub repo.

(B) 3D Stereotaxis Clamp

A lab has one 6mm vertical rod on their stereotaxic surgery table. They need to mount another 6mm rod 1" center-to-center from the mounted rod to roughly position an implant over the skull of a mouse. Assume both rods are 100mm in length.

1. Create two different clamp designs to achieve the intent.
2. Describe how you would manufacture a prototype clamp, and then, how to scale production to 100 parts.
3. Provide a Bill of Materials (BOM; with cost) and brief instructions for someone to recreate and deploy the clamps in their lab.

(C) Post-mortem Database Tool

Ideas and projects fail all the time, and a "post-mortem" is a good exercise to reflect and document why. You are tasked with creating a tool to organize these failures so we can track and search them.

1. Develop a strategic plan (<1 page, words only) for how you would create and deploy this. You may use any platform you would like or design your own.
2. Take initiative on your strategic plan and provide evidence (e.g., screenshots, link/s, code, etc.) for its basic functionality.
3. Use your new tool, or simply provide in writing, a post-mortem analysis of a failed idea, invention, or business concept from your career (<1 page).
4. Indicate what tools you used in your development (IDEs, AI, etc.).